

Aangan — Buy List

Nathaniel School of Music · house + studio Raspberry Pi system · rev 2 · every item is a clickable link

CORE buy this — the build needs it **CHEAPER ALT** blue rows — swap to save money, same job

OPTIONAL purple rows — nice-to-have / go-for-broke / later phase. Do NOT buy without asking Jason

How to use this: buy **BOTH projects now, as two SEPARATE orders** (one cart/invoice each) so the projects stay independent. Buy CORE items only, unless Jason approves an ALT/OPTIONAL row. SP Road first where marked — cheaper than Amazon and the school GSTIN gets input-tax credit. Before the Silverline trip WhatsApp **+91 70909 39819** to confirm stock (139/5 VT Complex, opp. Karthik Plaza, Nagarathpete, SP Road). Prices are estimates for budgeting.

Already owned — do NOT rebuy (₹0): Raspberry Pi 5 8GB · Pianoteq Pro licence · ESP32 dev boards x6 · ESP32-CAM x1 (assigned: main-entrance door-frame camera) · soil-moisture sensor x1 (balcony planter) · ultrasonic HC-SR04 x1 (overhead water tank) · buzzer alarm modules x4 (node sirens) · WS2812B LED strip 5m · INMP441 I2S mic · DHT22 x1 · PIR x4 · MQ-2 x1 · MC-38 reed sensors (count unclear — see red box) · water-leak sensors (count unclear — see red box) · jumper wires (only ~6 loose) · PCBs/enclosures · acoustic weatherstrip · old iPad (3rd panel) · certified smoke + LPG alarms · USB A→B keyboard cable · XLR cables · USB-C 100W cables · USB SSD · gaffer + velcro · studio UPS power backup.

⚠ **COUNT BEFORE ORDERING — the two inventories disagree:** one list says **9 reed door sensors and 6 water-leak sensors** are owned; the sourcing matrix says **zero reeds and 1 leak sensor**. Physically count both boxes first. Then: reeds — the house needs **12** (4 double doors get 1 each at the meeting stiles + front door + bedrooms/balconies), buy only the shortfall (~₹50 each). Leak sensors — the house needs **10** (4–6 sinks, stove lines, 2 bathrooms, balcony tap), buy only the shortfall (~₹80 each).

PROJECT 1 — Studio & House Pi ORDER NOW · Silverline visit + one robu.in/SP-Road run

A · THE BRAIN + PANEL DRIVERS (SILVERLINE)

ITEM	QTY	EST.	LINK
Raspberry Pi 5 — 8GB CORE the House brain — 8GB gives headroom for the doorbell-camera person detection later	1	₹8,000	silverline — Pi 5 8GB
Raspberry Pi 4 — 4GB CHEAPER ALT the matrix's pick, saves ₹2,500 — runs Home Assistant fine, but no headroom for camera AI later. Ask Jason before swapping.	(1)	₹5,500	silverline — Pi 4
Official Raspberry Pi 27W USB-C PSU CORE	1	₹1,300	silverline — 27W PSU
Raspberry Pi Zero 2 W CORE drives the two door screens	2	₹1,800 ea	silverline · robu fallback
5V/2.5A micro-USB PSU for the Zeros CORE	2	₹500 ea	silverline — PSU

B · THE TWO DOOR SCREENS

ITEM	QTY	EST.	LINK
Waveshare 10.1" HD (G), 1920×1200, toughened glass CORE FRONT DOOR — the OTP/visitor display, the one screen guests actually read up close	1	₹14,000	robu — Waveshare (G)
Waveshare 10.1" (E), 1024×600, optical bonding CORE STUDIO DOOR — a big ON-AIR sign read from a distance doesn't need full-HD. Saves ₹7,500 vs a second (G).	1	₹6,500	robu — Waveshare (E)
Upgrade: second Waveshare (G) for the studio door OPTIONAL only if Jason wants both doors full-HD	(1)	+₹7,500	same (G) link
Mini-HDMI → HDMI cable, 1.5m CORE Pi Zero uses MINI-HDMI — not micro	2	₹250 ea	robu — mini-HDMI

C · DOORBELL CAMERA — THE AFFORDABLE SOLUTION

ITEM	QTY	EST.	LINK
v1 = the owned ESP32-CAM (₹0) CORE already assigned to the main-entrance door frame in the sourcing matrix — snapshots to the app + panels. Buy only a momentary doorbell push button.	1 btn	₹30	robu — push button
v2 = TP-Link Tapo C210 (3MP, local RTSP) CHEAPER ALT to Reolink a real, sharp camera feed into our system for a quarter of the Reolink's price — the sensible upgrade when v1 feels grainy	(1)	₹2,800	amazon — Tapo C210
Reolink Video Doorbell PoE 2K + injector OPTIONAL · go-for-broke the premium route — only with Jason's explicit OK	(1)	₹12,000	amazon — Reolink

D · SENSORS + NODE PROTOTYPING (ONE ROBU/SP ROAD RUN)

ITEM	QTY	EST.	LINK
MC-38 reed door sensors CORE only the SHORTFALL after counting (need 12 total — see red box)	to 12	₹50 ea	robu — MC-38
Water-leak sensor modules CORE only the SHORTFALL after counting (need 10 total)	to 10	₹80 ea	robu — leak sensor
DHT22 climate sensors CORE bedrooms ×2, studio bed, residence hall (1 already owned = music room)	4	₹290 ea	robu — DHT22
MQ-6 LPG gas sensor modules CORE one per stove point; LPG-specific (owned MQ-2 becomes a spare/logger). Certified LPG alarm stays PRIMARY.	4	₹120 ea	robu — MQ-6
IR flame sensor modules CORE	4	₹70 ea	robu — flame
SW-420 vibration modules CORE perimeter + washing-machine-done	5	₹70 ea	robu — SW-420
HLK-LD2410 mmWave presence CORE	1	₹350	robu — LD2410

DFRobot SEN0232 sound level meter CORE STUDIO only — this is the meter that gates recording. The teaching room uses the owned INMP441 for now.	1	₹3,500	robu — SEN0232
Second SEN0232 for the teaching room OPTIONAL upgrade later if the INMP441 pilot isn't accurate enough	(1)	₹3,500	same link
JSN-SR04T waterproof ultrasonic CORE SUMP tank only — the owned HC-SR04 covers the overhead tank	1	₹450	robu — JSN-SR04T
HX711 + 50kg load cells (set) CORE LPG cylinder scale	1	₹400	robu — HX711
Jumper wire ribbon pack (120 pc) CORE only ~6 loose wires owned — nodes can't be built without this	1	₹350	robu — jumpers
Solderless breadboards CORE	3	₹150 ea	robu — breadboards
5V USB wall chargers + cables (node power) CORE one per ESP32 node — SP Road/local is cheapest	6	₹200 ea	robu — 5V adapters
SanDisk Extreme A2 microSD 64GB CORE House Pi + 2 panel Zeros	3	₹700 ea	robu — SanDisk
25A optoisolated relays (geyser switching) OPTIONAL · later phase only when the electrician does the geyser work — he fits them	(4)	₹450 ea	robu — 25A relay

NOT intern items (the electrician sources these): latching panic switches ×4, NC loop cable, battery-backed sounder, volt-free relay modules, motor contactor + float switches. A **low-voltage / fire-alarm technician** does this wiring.

PROJECT 2 — Music Pi (the piano rig) ORDER NOW · separate cart & invoice · uses the Pi 5 already owned

A · THE AUDIO CORE

ITEM	QTY	EST.	LINK
Raspberry Pi DAC Pro (IQAUDIO DAC Pro — PCM5242) CORE THE sound board · powered by the Pi, no soldering	1	₹2,500	robu — DAC Pro · search
XLR Board for IQAUDIO DAC Pro CORE balanced XLR sockets · ask robu first if they can order it in; otherwise import	1	₹2,000 + ship	robu — ask · pishop.us — import
Fallback: Behringer DI400P passive DI x2 ONLY IF the XLR board can't be sourced DAC Pro RCA → DI → balanced XLR	(2)	₹1,000 ea	amazon — DI400P
6-pin female header (2.54mm) CORE	2	₹30	robu — header

only if the DAC Pro ships with bare P7/P9 pads — ask the seller which revision

B · CONTROL, COOLING, POWER, STORAGE

ITEM	QTY	EST.	LINK
Raspberry Pi Touch Display 2 (7") CORE on-rig screen for switching pianos	1	₹5,500	silverline — Touch Display 2
Official Pi 5 Active Cooler CORE fit BEFORE the DAC goes on	1	₹500	silverline · robu
Official 27W USB-C PSU CORE	1	₹1,300	silverline — 27W PSU
SanDisk Extreme A2 microSD 64GB CORE one runs the rig, one is the cloned spare in the case	2	₹700 ea	amazon — SanDisk
100W USB-C PD power bank OPTIONAL · stage only the studio has UPS + mains — this is only for gigs with no wall socket	(1)	₹10,000	amazon — PD bank

C · MIDI IN EVERY FORM (USB IS NATIVE — THE OWNED A→B CABLE COVERS TODAY'S KEYBOARD)

ITEM	QTY	EST.	LINK
Roland UM-ONE mk2 USB-MIDI interface OPTIONAL only needed to connect 5-pin DIN instruments — the current keyboard is USB	(1)	₹4,000	amazon — UM-ONE
CME WIDI Master (wireless MIDI) OPTIONAL · go-for-broke Bluetooth-MIDI keyboards pair to the Pi for FREE — buy WIDI only for wireless DIN gear	(1)	₹9,000	amazon — WIDI Master

D · TRANSPORT

ITEM	QTY	EST.	LINK
Hard foam-lined case OPTIONAL · buy after assembly the stack must be measured first — the XLR board sits above the Pi's ports, normal cases don't close. Cables already owned.	(1)	₹3,000	measure, then buy

Aangan · Nathaniel School of Music · rev 2 (29 Jul 2026) — reconciled with the Master Sourcing Matrix · **CORE totals: Project 1 ≈ ₹47k** (₹20.5k of it is the two door screens; sensors themselves ≈ ₹8k) · **Project 2 ≈ ₹13.5k** · optional/go-for-broke rows add ₹40k+ ONLY if approved · full builds: “Aangan — Install Guide” + PIANO_PI_BUILD_PLAN.html in the aangan repo.